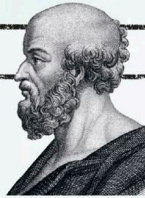


ERATOSTHENES OF CYRENE

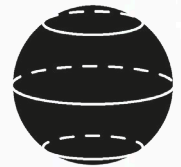
Born in Cyrene in North Africa, Eratosthenes became chief librarian at the library of Alexandria in Egypt, the greatest center of learning in the ancient world, at the age of 40.



Eratosthenes (276–194 BCE) drew maps of the world, calculated the distance from the Earth to the Sun, devised a reliable method for finding prime numbers, and coined the term “geography.” He was most renowned for accurately calculating the Earth’s circumference. He measured the angle of a shadow at Alexandria as 7.2° , while 500 miles (800 km) away at Swenet, the Sun was directly overhead. As 7.2° is one-fiftieth of the circumference of a circle (360°), he reasoned that the two points must be one-fiftieth of Earth’s circumference apart. He multiplied that distance by 50 to get 25,000 miles (40,000 km)—an error of less than 2 percent.



**PROVED
A SPHERE'S
VOLUME IS
TWO-
THIRDS
THAT OF A
CYLINDER
ENCASING
IT EXACTLY**



**SYRACUSIA
WEIGHED
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TONS**

**FOUND VALUE
OF π (PI)
TO BE
C.3.142**